

MATRICES

(1) trace of a matrix = sum of principal diagonal entries of a square matrix.

(2) transpose:

(i) $(A^T)^T = A$

(ii) $(A+B)^T = A^T + B^T$

(iii) $(AB)^T = B^T \cdot A^T$

(3) inverse of a matrix: $A^{-1} = \frac{1}{|A|} \cdot \text{adj}(A)$

(i) $(A^{-1})^{-1} = A$

(ii) $A \cdot A^{-1} = I$

(iii) $|A^{-1}| = 1/|A|$

(iv) $(AB)^{-1} = B^{-1} \cdot A^{-1}$

(v) $(A+B)^{-1} \neq A^{-1} + B^{-1}$

(4) determinant:

(i) $|A| = |A^T|$

(ii) $\det(AB) = \det(A) \cdot \det(B) = \det(B) \cdot \det(A) = \det(BA)$

(iii) $\det(A \pm B) \neq \det(A) \pm \det(B)$

(5) conjugate of a matrix: $(\bar{A})$

(i) $(\bar{\bar{A}}) = A$

(ii) $(\overline{A+B}) = \bar{A} + \bar{B}$

(iii) $(\overline{A \cdot B}) = \bar{A} \cdot \bar{B}$

(6) transjugate of a matrix: (A^0)

(i) $(A^0)^0 = A$

(ii) $(A+B)^0 = A^0 + B^0$

(iii) $(AB)^0 = B^0 \cdot A^0$

(7) symmetric matrix: $A^T = A$

• diagonal $\rightarrow$ any no.

• symmetric posn nos. are same.

(8) skew-symmetric matrix: $A^T = -A$

• diagonal $\rightarrow$ zero

• symmetric posn nos. are equal, opp. in sign

(9) hermitian matrix: $A^0 = A$

• diagonal elements are real.

(10) skew hermitian matrix: $A^0 = -A$

• diagonal elements are zero or imaginary

(11) unique representation of matrices:

matrix	expressed as	unique representation
square	$S + SS$	$A = \frac{1}{2}(A+A^T) + \frac{1}{2}(A-A^T)$
square	$H + SH$	$A = \frac{1}{2}(A+A^0) + \frac{1}{2}(A-A^0)$
square	$P + iQ$, P, Q are H.	$A = \frac{1}{2}(A+A^0) + i\left[\frac{1}{2i}(A-A^0)\right]$
Hermitian	$P + iQ$, P is real symm, Q is real skew symm	$A = \frac{1}{2}(A+\bar{A}) + i\left[\frac{1}{2i}(A-\bar{A})\right]$
SH.	$P + iQ$, Q is real ss, P is real sym	$A = \frac{1}{2}(A+\bar{A}) + i\left[\frac{1}{2i}(A-\bar{A})\right]$

(12) orthogonal matrix: $A \cdot A^T = A^T \cdot A = I$

(i) $|A| = \pm 1$ (if A is orthogonal)

(ii) $A^{-1} = A^T$

(13) unitary matrix: $A \cdot A^0 = A^0 \cdot A = I$

(i) $|A| = 1$

(ii) $A^{-1} = A^0$

(14) echelon form of a matrix:

no of zeros down the matrix should be increasing

(15) rank of a matrix:

(i) matrix of order $m \times n$, $1 \leq r(A) \leq \min(m, n)$

(ii) rank of non-sing. matrix = its order.

(iii) $r(A) = r(A^T)$

(iv) $r(AB) \leq r(A)$ and $r(AB) \leq r(B)$

$\therefore r(AB) \leq \min\{r(A), r(B)\}$

(v) rank of A = rank of (kA) , $k \rightarrow$ any scalar.

(vi) if $A_{n \times n}$ is non-singular ($|A| \neq 0$), rank of $A = n$

$r(A^2) = n$

(16) normal form of a matrix: $\begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}$

NOTE: rank of A = rank of normal form = r .

(17) cases for solution: (where $AX=B$, B is non-zero)

(i) consistent: (rank of A = rank of $[A:B]$)

(a) unique solution.

(b) infinite solutions

(ii) inconsistent:

rank $A \neq$ rank $[A, B] \rightarrow$ No $\rightarrow$ Inconsistent

yes

rank $A(r) = n \rightarrow$ Yes $\rightarrow$ unique soln.

No, $r < n$

infinitely many solutions.

when infinite solutions, $n-r$ (no. of var-rank) is no. of free variables

(18) In $AX=B$, $B=0$, homogeneous
 $B \neq 0$, non-homogeneous

(refer (17) for non-homo)

(19) homogeneous: ($AX=0$)

$x=0$ is $\rightarrow$ trivial or zero-solution.

$x \neq 0 \rightarrow$ non-trivial solution.

• rank of A (determine)

case I: $r(A) = r = n$, unique soln, zero-soln or trivial soln.

case II: $r(A) = r < n$, many non-trivial soln.
no. of independent solns, = $n-r$

AM-I

(1) division of complex nos: multiply numerator & denominator by conjugate of denominator

rank $A(\lambda) = n \xrightarrow{\text{yes}}$ unique (only trivial) soln.

↓ No. , then $r < n$. $\{r < n, \text{ if } |A| = 0\}$

infinitely many non-trivial solns

Note: In homo system of linear eqns,
no. of eq. (m) < no. of variables (n),
then, system will have infinitely many solns.
[converse, not true]

(20) linearly dependent vectors:

if $k_1 \neq 0$,

$$\lambda_1 = \mu_2 x_2 + \mu_3 x_3 + \dots + \mu_m x_m.$$

$$\mu = \frac{-k_i}{k_1} \quad (i=2,3,\dots,m)$$

if two set of vectors $x_1, x_2, \dots, x_m$ are linearly dependent, then any one vector can be expressed as linear combination of other vectors.

(21) Linearly independent vectors:

if $k_1x_1 + k_2x_2 + \dots + k_mx_m = 0$,

$$k_j = 0 \text{ for all } j = 1, 2, \dots, m.$$

$\therefore x_1, x_2, x_3, \dots$ are said to be linearly independent.

NOTE: If no. of components is less than no. of vectors, then vectors are always linearly dependent.

Rank is equal to no. of independent rows and columns.

(22) Gauss-Jacobi's method:

- use previous iteration values
- strictly or irreducibly diagonally dominant.
- initials not given, assume $x=0, y=0, z=0$.

(23) Gauss-Seidel Method:

- use latest two values,
- strictly or irreducibly diagonally dominant
- initials not given, assume $x=0$, $y=0$, $z=0$.

↓
29.

$$\sigma_{\text{sup}} \leftarrow \frac{\partial V}{\partial \sigma_{\text{sup}}} \quad \sigma_{\text{inf}} \leftarrow \frac{\partial V}{\partial \sigma_{\text{inf}}}$$
 $\sigma > 8,046$

continued

(1947-1948) 1-10 10/10/47 10/10/47

13/10 AX=B, B=0, B=XA

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(continued on next page)

$$(0.001 \times 10^3) = 2009 \text{ cm}^3 \text{ (approx.)}$$

Conductor - 3 ms to 10 ms $\leftarrow$ 4.0 x

Count of A (1965-1970)

$\forall \alpha \in A, \exists \beta \in B : \alpha = c + (\beta)^2$

for $\alpha \in \mathbb{R}$ and $\beta \in \mathbb{R}$ with $\alpha + \beta = 1$. The function f is called a β - α -B-spline of order k if it is a linear combination of the k B-splines of order k with support on $[\alpha, \beta]$. The function f is called a β - α -B-spline of order k if it is a linear combination of the k B-splines of order k with support on $[\alpha, \beta]$.

10. 11. 1940

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$$(I - A)A^{-1} = (I + A)^{-1} = A \quad 22 + 2$$

$$(\partial A - A) \frac{1}{L} + (\partial A + A) \frac{1}{L} = A' \quad H_2 + H$$

$$\left[\frac{(G_A + A)}{(\lambda)} \right]_i + \left[\frac{(G_A + A)}{(\lambda)} \right]_i = A \quad \text{H. 3100, 1}$$

$$[(\bar{A}-1) \frac{1}{2}]! + (\bar{A}+1)! \quad A = \frac{1}{2}, 1, 3/2, 2, \dots$$

$$\left[(\bar{A}-A) \frac{1}{\sqrt{N}} \right] + (\bar{A}+A) \frac{1}{\sqrt{N}} = A \quad \text{при } \theta = 0, \text{ т.е.}$$

mythology

HOMOGENEOUS FUNCTIONS

- (1) for two variables: $u = f(x, y)$ is called homo. function of degree n if $u = x^n \cdot f\left(\frac{y}{x}\right)$, $n \rightarrow \text{real no.}$
- (2) for three variables: $u = f(x, y, z)$ is homo. function of degree n if $u = x^n \cdot f\left(\frac{y}{x}, \frac{z}{x}\right)$

rule:

- (i) $x = xt, y = yt, z = zt$
 (ii) find $f(xt, yt, zt)$
 (iii) if $f(xt, yt, zt) = t^n \cdot f(x, y, z)$, then $u = f(x, y, z)$ is called homo. func. of degree n .
- (3) Euler's theorem:

- (i) for function of two variables:

If $u = f(x, y)$ is homo. func. of degree n , then,

$$x \cdot \frac{\partial u}{\partial x} + y \cdot \frac{\partial u}{\partial y} = nu$$

- (ii) for function of three variables:

If $u = f(x, y, z)$, we get,

$$x \cdot \frac{\partial u}{\partial x} + y \cdot \frac{\partial u}{\partial y} + z \cdot \frac{\partial u}{\partial z} = nu$$

- (4) Corollary 1:

(i) If $u = f(x, y)$ is homo func. of two variables x and y of degree n , then,

$$x^2 \cdot \frac{\partial^2 u}{\partial x^2} + 2xy \cdot \frac{\partial^2 u}{\partial x \partial y} + y^2 \cdot \frac{\partial^2 u}{\partial y^2} = n(n-1)u$$

- (ii) for $u = f(x, y, z)$,

$$x^2 \cdot \frac{\partial^2 u}{\partial x^2} + y^2 \cdot \frac{\partial^2 u}{\partial y^2} + z^2 \cdot \frac{\partial^2 u}{\partial z^2} + 2xy \cdot \frac{\partial^2 u}{\partial x \partial y} + 2yz \cdot \frac{\partial^2 u}{\partial y \partial z} + 2zx \cdot \frac{\partial^2 u}{\partial z \partial x} = n(n-1)u$$

- (5) Corollary 2:

If $f(u)$ is homo. func. of degree n , then,

$$x \cdot \frac{\partial u}{\partial x} + y \cdot \frac{\partial u}{\partial y} = n \cdot \frac{f(u)}{f'(u)}$$

{ u is not homo. }

for three variables,

$$x \cdot \frac{\partial u}{\partial x} + y \cdot \frac{\partial u}{\partial y} + z \cdot \frac{\partial u}{\partial z} = n \frac{f(u)}{f'(u)}$$

- (6) Corollary 3: $f(u)$ homo function,

$$x^2 \cdot u_{xx} + y^2 \cdot u_{yy} + z^2 \cdot u_{zz} + 2xy \cdot u_{xy} + 2yz \cdot u_{yz} + 2zx \cdot u_{zx} = g(u) [g'(u) - 1]$$

where,

$$g(u) = \frac{n \cdot f(u)}{f'(u)}$$

for three variables,

$$x^2 \cdot u_{xx} + y^2 \cdot u_{yy} + z^2 \cdot u_{zz} + 2xy \cdot u_{xy} + 2yz \cdot u_{yz} + 2zx \cdot u_{zx} = g(u) [g'(u) - 1]$$

PARTIAL DIFFERENTIATION AND ITS APPLICATION

$$(1) \frac{\partial^2 u}{\partial x \partial y} = \frac{\partial^2 u}{\partial y \partial x}$$

$$\text{Formula: } (x^2 + y^2 + z^2 - yz - zx - xy)(x + y + z) - x^3 - y^3 - z^3 - 3xyz$$

- (2) composite function:

$$x = \phi(t), y = \psi(t),$$

$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt}$$

- (3) maxima and minima:

method:

- (i) $f(x, y)$, $\frac{\partial f}{\partial x} = 0, \frac{\partial f}{\partial y} = 0$, for x and y simultaneously

- (ii) put the points and get the points. (a, b) is stationary pt.

$$(iii) \frac{\partial^2 f}{\partial x^2} = r, \frac{\partial^2 f}{\partial y \partial x} = s, \frac{\partial^2 f}{\partial y^2} = t$$

$$\therefore rt - s^2 > 0, r < 0 \text{ or } t < 0, f(x, y) \text{ max at } (a, b)$$

$$rt - s^2 > 0, r > 0 \text{ or } t > 0, \text{ min.}$$

$$rt - s^2 < 0, f(x, y) \text{ neither max or min at } (a, b)$$

[saddle point]

$$rt - s^2 = 0, \text{ test fails}$$

- (4) Jacobian of two and three independent variables:

$$u = f(x, y), v = g(x, y)$$

$$J = \frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix}$$

$$J^{-1} = \frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix}$$

$$J \cdot J^{-1} = 1$$

$$J \left(\frac{u, v, w}{x, y, z} \right) = \frac{\partial(u, v, w)}{\partial(x, y, z)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{vmatrix}$$

COMPLEX NUMBERS

(1) division of complex nos: multiply and divide by conjugate of denominator

(2) $|z| = \text{mod}(z) = r = \sqrt{x^2 + y^2}$

(3) $\arg z = \theta = \tan^{-1}\left(\frac{y}{x}\right)$

(4) $y = r \sin \theta, x = r \cos \theta$

(5) $i = \sqrt{-1}, i^2 = -1, i^3 = -i, i^4 = 1$

(6) cartesian form: $z = x + iy$

polar form: $z = r(\cos \theta + i \sin \theta)$

exponential form: $z = r.e^{i\theta}$

(7) properties of a complex no.

(i) $\text{Re}(z) = x = \frac{1}{2}(z + \bar{z})$

(ii) $\text{Im}(z) = y = \frac{1}{2i}(z - \bar{z})$

(iii) $(z_1 + z_2) = \bar{z}_1 + \bar{z}_2$

(iv) $(z_1 \cdot z_2) = \bar{z}_1 \cdot \bar{z}_2$

(v) $\left(\frac{z_1}{z_2}\right) = \frac{\bar{z}_2}{\bar{z}_1}$

(vi) $z \cdot \bar{z} = |z|^2 = |\bar{z}|^2 \quad \{ |z| = |\bar{z}| = \sqrt{x^2 + y^2} \}$

(vii) $|z_1 z_2| = |z_1| |z_2|$

$\arg(z_1 z_2) = \arg(z_1) + \arg(z_2)$

(viii) $\left| \frac{z_1}{z_2} \right| = \frac{|z_1|}{|z_2|}$

$\arg\left(\frac{z_1}{z_2}\right) = \arg(z_1) - \arg(z_2)$

x	y	amplitude
$x > 0$	$y > 0$	$\text{amp}(z) = 0$ [P(x,y) lies on +ve side of x-axis]
$x < 0$	$y = 0$	$\text{amp}(z) = \pi$ [P(x,y) lies on -ve side of x-axis]
$x = 0$	$y > 0$	$\text{amp}(z) = \frac{\pi}{2}$ [P(x,y) lies on +ve side of y-axis]
$x = 0$	$y < 0$	$\text{amp}(z) = \frac{3\pi}{2}$ [P(x,y) lies on -ve side of y-axis]

(9) $\cos \theta = \frac{x}{r} = \frac{x}{|z|}; \sin \theta = \frac{y}{r} = \frac{y}{|z|}$

x	y	range of θ
$x > 0$	$y > 0$	$0 < \theta < \frac{\pi}{2}$ (1st Quad.)
$x < 0$	$y > 0$	$\frac{\pi}{2} < \theta < \pi$ (2nd Quad.)
$x < 0$	$y < 0$	$\pi < \theta < \frac{3\pi}{2}$ (3rd Quad.)
$x > 0$	$y < 0$	$\frac{3\pi}{2} < \theta < 2\pi$ (4th Quad.)

* $(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$

* $(a-b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$

* $(z_1 + z_2)^2 = (z_1 - z_2)^2 - 4z_1 z_2$

* if z_1 and z_2 are roots of a quad. eqn. then eqn will be,

$x^2 - (\text{sum of roots})x + (\text{products of roots}) = 0$

Formula for roots:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

* $1 + \cos 2\theta = 2\cos^2 \theta$
 $\sin 2\theta = 2\sin \theta \cdot \cos \theta$

(10) De-Moivre's theorem:

$(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta$

$z = \cos \theta + i \sin \theta$

$\therefore z^{-1} = \frac{1}{z} = \cos \theta - i \sin \theta$

$(\cos \theta - i \sin \theta)^n = \cos n\theta - i \sin n\theta$

$(\sin \theta + i \cos \theta)^n = \left(\cos n\left(\frac{\pi}{2} - \theta\right) + i \sin n\left(\frac{\pi}{2} - \theta\right) \right)$

* $(x+y+z)^3 = x^3 + y^3 + z^3 + 3(x+y+z)(xy+yz+zx) - 3xyz$

* $S_n = \frac{n}{2} [2a + (n-1)d] \rightarrow a \rightarrow \text{initial term } \{A.P.\}$
 $d \rightarrow \text{difference}$

(11) roots of algebraic equations:

$z^n = \text{complex no.}$

Steps:

(i) $z^n = r(\cos \theta + i \sin \theta)$

(ii) $z^n = r(\cos(\theta + 2k\pi) + i \sin(\theta + 2k\pi))$

(iii) $z = r^{1/n} (\cos(\frac{\theta + 2k\pi}{n}) + i \sin(\frac{\theta + 2k\pi}{n}))^{1/n}$

(iv) $z = r^{1/n} \left[\cos\left(\frac{\theta + 2k\pi}{n}\right) + i \sin\left(\frac{\theta + 2k\pi}{n}\right) \right]$

$\{ r = \sqrt{x^2 + y^2} \}$ $k = 0, 1, 2, \dots, n-1$

$\left\{ \tan \theta = \frac{y}{x} \rightarrow \theta = \tan^{-1}\left(\frac{y}{x}\right) \right\}$

* $(x^5 - 1) = (x-1)(x^4 + x^3 + x^2 + x + 1)$

power decrease is continuous

$\therefore$ take one higher power

* $\sin(A-B) = \sin A \cdot \cos B - \cos A \cdot \sin B$

(12) hyperbolic functions:

$\cosh x = \frac{e^x + e^{-x}}{2}$

$\sinh x = \frac{e^x - e^{-x}}{2}$

$\cos z = \frac{e^{iz} + e^{-iz}}{2}$

$\sin z = \frac{e^{iz} - e^{-iz}}{2}$

$\tanh x = \frac{e^x - e^{-x}}{e^x + e^{-x}}$

imp relation b/w circular and hyperbolic functions:

$$\sinh z = -i \sin iz$$

$$\cosh z = \cos iz$$

$$\sin z = -i \sinh iz$$

$$\cos z = \cosh iz$$

$$\int \cosh x \cdot dx = \sinh x$$

$$\int \sinh x \cdot dx = \cosh x$$

$$\int \operatorname{sech}^2 x \cdot dx = \tanh x$$

{ multiplication of sin results in
-ve sign }

$$\frac{d \sinh x}{dx} = \cosh x$$

$$\frac{d \cosh x}{dx} = \sinh x$$

$$\frac{d \tanh x}{dx} = \operatorname{sech}^2 x$$

$$* \sin(a+b) = \sin a \cos b + \cos a \sin b$$

$$* n C_r = \frac{n!}{(n-r)! r!}$$

$$* (a+b)^n = a^n + n C_1 a^{n-1} b + n C_2 a^{n-2} b^2 + \dots + b^n$$

$$* 1 + \cos^2 \theta = \cot^2 \theta$$

$$* \tan(a+b) = \frac{\tan a + \tan b}{1 - \tan a \cdot \tan b}$$

For tan inverse Questions

$$\text{eg. let } \tan^{-1}(e^{i\theta}) = x + iy.$$

$$e^{i\theta} = \tan(x + iy)$$

$$\cos \theta + i \sin \theta = \tan(x + iy)$$

$$\tan 2x = \tan(x + iy + x - iy) \Rightarrow \tan 2x = \frac{\tan(x + iy) + \tan(x - iy)}{1 - \tan(x + iy) \tan(x - iy)}$$

$$\tan(2iy) = \frac{\tan(x + iy) + \tan(x - iy)}{1 - \tan(x + iy) \tan(x - iy)}$$

$$\tan(2iy) = \frac{\tan(x + iy) + \tan(x - iy)}{1 - \tan(x + iy) \tan(x - iy)}$$

$$\left\{ \begin{array}{l} \tan(x + iy) = \alpha + i\beta \\ \tan(x - iy) = \alpha - i\beta \end{array} \right\}$$

$$* \sinh^{-1} x = y = \log(x + \sqrt{x^2 + 1})$$

$$* \cosh^{-1} x = \log(x + \sqrt{x^2 - 1})$$

$$* \tanh^{-1} x = \frac{1}{2} \log \left(\frac{1+x}{1-x} \right)$$

$$* \cos(a+b) = \cos a \cos b - \sin a \sin b$$

(13) logarithms of a complex no:

$$\log(x + iy) = \frac{1}{2} \log(x^2 + y^2) + i \tan^{-1} \left(\frac{y}{x} \right)$$

Principal value of $\log z$ $-\pi \leq \theta \leq \pi$

$$\operatorname{Log}(x + iy) = \frac{1}{2} \log(x^2 + y^2) + i \left(\tan^{-1} \left(\frac{y}{x} \right) + 2n\pi \right)$$

general value of $\log(x + iy)$

$$\begin{aligned} \log z &= \log(x + iy) = \log r + i\theta \\ \log z &= \log(x + iy) = \log r + i\theta \\ \log z &= \log(x + iy) = \log r + i\theta \end{aligned}$$

division of complex no: multiply and divide by
conjugate of denominator

$$\frac{z_1}{z_2} = \frac{z_1 \cdot \overline{z_2}}{z_2 \cdot \overline{z_2}} = \frac{z_1 \cdot \overline{z_2}}{|z_2|^2}$$

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